

MultiPlanetary Exchange System — Participant Brief

Released October 2, 2026

1. The commission

You have 40 hours, from 6:00 PM Eastern on Friday, October 2, 2026 to 10:00 AM Eastern on Sunday, October 4, 2026, to design and document a working MultiPlanetary Exchange System: the markets, clearing, settlement, and supporting institutions that keep running while the planets and communication relays move. The story is set on 22 September 2126, but every planet starts where the real solar system is on 22 September 2026.

Investors on Earth finance construction on Mars. Asteroid producers and manufacturers sign long-term contracts. Your exchange must let people at all nine settlements invest, trade, transfer assets, and enter financial obligations, and it must keep working for as long as the modeled civilization runs. The 40 hours is your deadline, and it is real clock time. It is not the market's lifetime.

Your design must be precise enough that another person can follow a transaction from its first instruction to its final outcome. State what you guarantee, the conditions under which you guarantee it, and what participants pay or give up.

There is no fixed assignment. You choose your own opening balance sheet (within the limits in Section 6) and your own test scenarios. The solar system, the network, and the communication rules are the same for every team, and those shared rules are what let judges compare designs fairly.

Scope

The scope covers equities, futures, options, bonds, currencies, lending, short positions, clearing, and settlement. You do not need to specify every contract. Explain a general framework, demonstrate it on your own scenarios (Section 6), and mark each rule as implemented or as an extension. A list of product names is not evidence of support.

Who can compete

Anyone may enter, subject to the hackathon-wide rules. Code is welcome but optional. Calculations, spreadsheets, worked timelines, and clearly written procedures can earn full credit. A reference orbit propagator and epoch test vectors are supplied so nobody has to spend the first hours debugging frame conventions. Polish and code volume earn no points. Code is optional for the design itself, but the 200-year scan and the moving-receiver calculations are impractical by hand, so expect to use code or a spreadsheet for E1 and E4.

What is the same for every team, and what you design

Same for every team	Designed by you
Geography, orbits, and the relay network	Market structure and who is responsible for what

Same for every team	Designed by you
Communication services, loss, capacity, and quotas	Contract rules, matching, and clearing
Physical limits and timing rules	Margin, valuation, and default procedures
Scoring and evidence requirements	Your opening balance sheet (within the limits in Section 6)
	Your test scenarios, including the incident you use to stress your design
	User access and operating procedures
	Where each institution is located (any of the nine settlements)

You cannot add network infrastructure or change its limits. No architecture is preferred. Centralized, federated, batch, and other designs can all score well if they accept the tradeoffs they create. You must identify your own design risks.

How to read this brief

Section 2 collects every fixed number in one place. If the text elsewhere disagrees with a table in Section 2, the table governs. Report the disagreement to the organizers (Section 11).

- Section 3: the moving solar system.
- Sections 4 and 5: communication, transport, and disruptions.
- Sections 6 and 7: your balance sheet and scenarios, the funding rules, and probability accounting.
- Section 8: the evidence you must submit.
- Section 9: continuing operation and deployment effects.
- Section 10: submission and scoring.
- Section 11: how clarifications are issued during the event.

Section 10 ends with a submission checklist. Section 2 also works as a glossary.

2. Terms and fixed parameters

Every number below is the same for all teams. Times are elapsed simulation time from the epoch unless stated otherwise.

Terms

Term	Meaning
Epoch, $t = 0$	2026-09-22 00:00:00 TDB. The story date (2126-09-22) is only a label. Do not move planetary positions forward by 100 years at $t = 0$. Elapsed model time starts at zero. Shifted epochs (E5) are the only place where positions are advanced from the epoch. The epoch has nothing to do with the hackathon dates: the event runs on the real calendar, and $t = 0$ is the start of the simulated solar system.
Hour h	Elapsed time since $t = 0$. One day is 86,400 s. One Julian year is 365.25 days.
NeoDollar (\$)	The unit of money in this brief, written with the \$ sign. It is an abstract unit of account: no bank issues it, it pays no interest, and no legal system stands behind it. Nobody can create new NeoDollars (see Section 7).
Settlement	One of nine sites: Mercury, Venus, Earth, Mars, Ceres, Jupiter, Saturn, Uranus, Neptune. Each giant-planet community is represented by its planetary gateway.
Gateway	A settlement's single access point to the network.
Node	A gateway or a relay (Relay A, Relay B).
Principal	One identity that holds accounts and sends messages. Each named account in your balance sheet is one principal, and so is each institution you charter. One participant or institution is one principal, however many locations or accounts it has. You may charter at most 12 institutions in total. Each institution has its own direct quota of 12 packets per rolling 24 h, for client communication only. Its backbone originations count against the shared 600.
Node ID	Used to order simultaneous arrivals: Mercury 1, Venus 2, Earth 3, Mars 4, Ceres 5, Jupiter 6, Saturn 7, Uranus 8, Neptune 9, Relay A 10, Relay B 11.
Operator	An institution allowed to use the backbone. Operators and their clearing or settlement services are the only backbone users.
Official inter-exchange coordination	Any message that creates a shared exchange record or settlement authority. Messages are classified by what they do, not by their label. Official coordination must use the backbone.
Client communication	Orders, instructions, and observations sent by clients, agents, or information sources. These use local access or the direct service.
Originated packet	A new packet created by the application, including new sessions and application resubmissions. Automatic retries and transport acknowledgments are not originated packets.
Conditional (no-loss) trace	A timeline that removes random packet loss only. Geometry, quotas, timers, queues, and maintenance still apply. It is not a delivery guarantee.
Scenario	One run of your design on your opening balance sheet, from hour 0 to a stated end (Section 6).
Reset	Restoring the full opening balance sheet for a separate run. Orbital phase is never reset.

Time and motion

Parameter	Value
Frame	Sun-centered, geometric, three-dimensional J2000 ecliptic, in AU
Propagation	Independent fixed Kepler ellipses from the supplied epoch elements
Solar exclusion	Opaque sphere of radius 0.10 AU. A photon path that crosses it is unavailable. The test uses the straight segment from the sender's position at emission to the receiver's position at arrival.
Times before t = 0	Positions before t = 0 use the same Kepler propagation run backward.
Relays A and B	Prograde circular heliocentric orbits of radius $\sqrt{8}$ AU in the reference plane, with epoch phases 45° and 135°
Relay period	$365.2568983 \times (\sqrt{8})^{1.5}$ days, about 1,737 days
Light time	Solve $t_a - t_e = 8.317 \text{ min/AU} \times r_{\text{receiver}}(t_a) - r_{\text{sender}}(t_e) $ for t_a , with positions in AU and times in minutes. Fixed-point iteration is fine, to a tolerance of 1 ms in t_a .
Launch timing	Serialization takes 1 s per launch, on every hop. Emission t_e is 1 s after the packet is ready to launch on that link. R_h starts at t_e and covers flight time only.
Relay processing	A relay adds 1 s between receiving a packet and making it ready to launch on the next link.
Local access	1 s each way at every gateway, with no loss. This stands in for all other occultation. It applies to client communication only: direct packets, and messages between a client and an institution at the same settlement. Each operator and its clearing or settlement services sit at their settlement's gateway, and a backbone launch starts at the gateway, so backbone launches pay no local access.

Communication

Parameter	Backbone	Direct
Loss per launch	$1 - \exp(-0.02 \text{ d})$	$1 - \exp(-0.08 \text{ d})$
Who may use it	Operators and their clearing or settlement services	Any principal, for client communication only
Origination quota	600 packets per rolling 24 h, shared by the whole system	12 packets per rolling 24 h per principal, launches at least 60 s apart
Capacity	1 packet per second per directed link, FIFO, 10,000-packet queue, newest dropped when full	None beyond the quota
Relay, retry, and receipt service	Yes (Section 5)	None

Here d is the actual photon path length in AU to the receiver's arrival position. Loss is applied anew to every launch, retry, receipt, and reply. A packet holds at most 1,024 bytes, including a 64-byte header, so the payload is at most 960 bytes. Encoding is fixed-width and uncompressed. Simultaneous arrivals are ordered by ascending sender node ID (Section 2, Terms), then by packet sequence number. The sender assigns sequence numbers in the order it creates packets.

Transport timers

Parameter	Value
Route	Simple (no repeated node), at most 3 backbone links, pinned for the session
Hop retry timer R_h	$2 \times$ calculated one-way flight time + 60 min; at most 4 launches
Endpoint retry timer R_e	$2 \times T_0 + 24$ h, where T_0 is the predicted empty-queue one-way route delay at enqueue; at most 4 attempts for SYN, SYN-ACK, and each data packet
Send window	64 unacknowledged data packets per peer connection
Packet lifetime	30 days from original creation, covering all hop and endpoint retries
Duplicate-ID retention	Packet lifetime plus 30 days after the last receipt
Session expiry	7 days without receiving a session packet, or on a declared reset. Session identities are never reused.

Disruptions

Parameter	Value
Scheduled maintenance (one-time, both directions)	B to Neptune during hours [2, 26). B to Ceres during hours [240, 264).
Additional incident, at most one per tested window	A 72 h gateway isolation, a 6 h forced-loss incident, or an endpoint reset, defined below. In the S2 stress run (Section 8) you choose the kind, the node, and the start time. Incidents are never announced in advance: participants learn of one only from what fails to arrive, except that a reset endpoint knows it was reset.
Forced-loss incident	Every launch on any link with the named node as an endpoint fails if its emission time t_e falls inside the window. Loss is decided at emission. Packets already in flight when the window opens are unaffected. Hop receipts, SYN, SYN-ACK, ACK, and data are launches like any other. Retries emitted inside the window fail; retries emitted after it are unaffected.
Gateway isolation	For 72 h, every launch to or from the named settlement fails if its emission time falls inside the window. This covers backbone launches from or to its gateway and direct packets to or from it. Local access at that settlement keeps working, so parties there can still talk to each other and to institutions there. Packets already in flight when the window opens are unaffected. All records are kept.

Parameter	Value
Endpoint reset	The named endpoint (an operator, a clearing or settlement service, or an agent) loses all of its session state at one moment: open sessions, send windows, retry timers, and unsent queues. It keeps its durable financial records and its packet-ID records. Every session it had is dead and must be replaced by a new handshake, which costs one quota packet. Packets already in flight are unaffected, but the endpoint ignores any that belong to a dead session.
Natural solar blockage	Recurrent, and in addition to the above. It comes from geometry and is not an incident.

Your balance sheet and evaluation

Parameter	Value
Opening balance sheet limits	At most \$500,000 (NeoDollars) and at most 5,000 shares in total, held across 4 to 10 named accounts located at 3 or more different settlements (Section 6). Institutions you charter start with nothing.
Shares	Shares of one or more companies you name (for example, Ares Habitat)
Orbital validation span	At least 200 Julian years
Shifted-epoch offsets	1, 10, and 100 Julian years, plus one epoch you choose

3. The moving solar system

Every body follows a fixed Kepler ellipse from the supplied epoch elements, so distances and contact windows change continuously. You must calculate them. No distance table or contact schedule is supplied. Freeze the elements, not the positions.

Supplied data

Alpha_Orbital_Data.zip contains orbital_elements.json, network_model.json, the nine original NASA/JPL Horizons responses, and a data dictionary. It also contains a reference Kepler propagator and the epoch check positions below. The archive is frozen: do not query a live ephemeris or switch ephemerides while comparing results.

Each body has a semi-major axis a , eccentricity e , inclination i , ascending node Ω , argument of perihelion ω , mean anomaly M_0 at the epoch, and mean motion n in degrees per day.

How to propagate

- Advance the mean anomaly: $M(t) = M_0 + n \times (\text{elapsed days})$, modulo 360° .
- Solve Kepler's equation $E - e \sin E = M$ for E .
- Compute orbital-plane coordinates $x' = a (\cos E - e)$ and $y' = a \sqrt{1 - e^2} \sin E$.
- Rotate into the J2000 ecliptic frame using ω , then i , then Ω .

Relays A and B are not in the archive's element list. Each moves on a prograde circle of radius $\sqrt{8}$ AU in the reference plane, at angle phase $+ 360^\circ \times t / \text{period}$. The Sun is at the origin, so a straight photon path is unavailable if its closest approach to the origin is under 0.10 AU.

This is a common, simplified model that can run for any length of simulated time. It does not claim that unperturbed ellipses predict the real solar system forever, and you should not treat it that way.

Epoch check positions

Your implementation should reproduce these to within 0.00001 AU at $t = 0$. Each body should also return to its starting point after its orbital period and keep its distance from the Sun between $a(1 - e)$ and $a(1 + e)$.

Body	x (AU)	y (AU)	z (AU)
Mercury	-0.213426	-0.410281	-0.013955
Venus	0.679067	-0.257950	-0.042725
Earth	1.003581	-0.023693	-0.000003
Mars	0.247476	1.526869	0.025929
Ceres	0.375786	2.653941	0.014791
Jupiter	-3.438179	4.038309	0.060149
Saturn	9.271221	1.717485	-0.398962
Uranus	8.962475	17.253919	-0.052129
Neptune	29.839098	1.351785	-0.715470
Relay A	2.000000	2.000000	0.000000
Relay B	-2.000000	2.000000	0.000000

Physical boundaries

- Bodies other than the Sun never block a path in the baseline. Their occultation, rotating ground stations, satellite handoffs, and planetary shadows are all folded into the 1 s lossless local access at each gateway.
- Nothing travels faster than light. There is no instant remote action and no physical courier shortcut.
- All parties share one time coordinate and have unforgeable identities. Clocks reveal nothing about remote events.
- Trust and faults. Hour-0 balances and account ownership are authoritative as you declare them. Clients may act strategically: they may try to spend a pledged asset twice, or lie about what they know. Operators follow your published rules, but they can crash, be reset, or be cut off from the network, and your design must handle that. Operators that break their own rules (malicious or colluding operators) are an optional extension, and you must label it as one. Your rules decide whether an owner can spend a pledged asset. If they cannot, say what stops them.

- Exchanges keep durable financial records.
- Higher-fidelity gravity or local access assumptions are allowed only as clearly labeled extensions. They never silently change the common baseline. Section 9 says how they are assessed.

4. Communication services

There are two ways to move a message between settlements: a reliable but scarce official backbone, and a cheap but lossy direct service. Each has its own users, loss law, and quota. Clients and information sources use the direct service or local access. Only exchange operators use the backbone.

The backbone

The backbone has 19 candidate two-way links: each of the nine gateways to Relay A, each of the nine gateways to Relay B, and Relay A to Relay B. There are no other backbone links, and you cannot add or move any.

A candidate link can carry a launch only when both of these hold at that moment:

- The straight photon path from the sender at emission to the receiver at arrival misses the 0.10 AU solar exclusion sphere.
- No scheduled or additional outage applies (Section 5).

Exchange-to-exchange coordination, including remote clearing and settlement, must use this backbone. Link lengths, visibility, and usefulness all change with time.

The direct service

Any two settlements with an unobstructed direct path can exchange packets without the backbone. There is no automatic relay, receipt, or retry. A sender learns no delivery result unless the receiver replies, and the reply is a separate flight at a later time. The direct service cannot carry official exchange-to-exchange coordination.

A direct packet takes 1 s of local access at the sending settlement, 1 s of serialization, the flight time from the light-time formula, and 1 s of local access at the receiving settlement. It has no relay processing and no queue. The same 0.10 AU solar-exclusion test applies to direct paths. Communication between two parties at the same settlement uses local access only: 1 s, no loss, and no quota. It is not a direct packet.

Loss and timing

For every launch, let d be the photon path length in AU to the receiver's arrival position.

Backbone loss probability: $p = 1 - \exp(-0.02 \times d)$. Direct loss probability: $p = 1 - \exp(-0.08 \times d)$. For example, at $d = 1$ AU the backbone loses a packet with probability 1.98% and the direct service with probability 7.69%.

- Each launch is an independent random trial, across links, packets, directions, and time.
- Loss applies once per launch on each link. There is no second loss event for the whole route.
- Receipts, acknowledgments, handshake messages, retries, and replies all use the same formula, recomputed from their own launch geometry.
- A lost packet is simply absent at its nominal arrival time. Undetected corruption does not exist.

- Order of events for one launch: 1 s serialization, emission at t_e , then flight for $8.317 \times \text{distance}$ minutes, solved as in Section 2.

Capacity and quotas

Item	Backbone	Direct
Who can originate	Operators and their clearing or settlement services	Any principal
Packet size	At most 1,024 bytes including headers	Same
Link capacity	1 packet per second per directed link, FIFO, 10,000-packet queue	None beyond the quota
Origination quota	600 per rolling 24 h for the whole system. Chartering more operators does not raise it.	12 per rolling 24 h per principal, at least 60 s between launches
Counts against the quota	Initial data, application control packets, the SYN of each new session (a replacement session after expiry or reset is a new session and costs one), and application resubmissions. "Application control packets" means application-level messages, not transport messages.	Every packet, including copies and acknowledgments
Exempt from the quota	Automatic retries and transport acknowledgments: SYN-ACK, the final ACK, hop receipts, and data acknowledgments. They still use link capacity and count in your communication totals.	Nothing
Quota shared with the other service?	No	No

- The rolling 24 h window is continuous. Session setup before hour 0 counts in the same window as later traffic, and in your communication totals.
- Larger messages use several packets. When a queue is full, the newest arrival is discarded.
- No free global feed, pooled anonymous capacity, extra terminal, or unlisted route exists. Local information is available immediately where it starts. Official publications travel in exchange sessions.

Classifying traffic

A message is classified by what it does. A message that creates a shared exchange record or settlement authority is official coordination and needs the backbone. You may not relabel an exchange reconciliation service as a private trader to reach the direct service. Client instructions and observations stay client communication.

A price observation, and any direct copy of it sent to a client or institution, is client communication and may use the direct service under the sender's own direct quota. A message that an operator or clearing service sends to another operator or ledger to create a shared exchange record or settlement authority is official coordination and must use the backbone, even if it carries or comes from an observation. This includes marks, margin calls, and settlement instructions, and it counts against the shared 600 quota. An institution may act locally on an observation it received directly, but that action becomes a shared record only through the backbone.

5. Transport, sessions, and disruptions

The backbone uses a simplified, TCP-inspired transport defined only by this section. It is not a full implementation of Internet TCP. Relays move bytes. Exchanges decide what those bytes mean financially. A transport acknowledgment means the bytes arrived. It never means a financial action was accepted.

Hop delivery through relays

Before a session starts, choose a simple route (no repeated node) of at most three backbone links. The route may pass through Relay A or Relay B. It is pinned for that session. At every hop:

- The forwarding node launches the packet and sets $R_h = 2 \times \text{its calculated one-way flight time} + 60 \text{ minutes}$.
- The receiving node stores the packet durably, queues a hop receipt on the reverse link at once, and forwards the first copy to the next hop.
- A duplicate copy triggers a new receipt but is not forwarded again.
- If no receipt arrives by R_h , retry. There are at most four launches in total, and each timer is recomputed from that launch's geometry.
- After the fourth timeout, abandon the hop transfer.

Hop receipts are one-shot: they do not ask for receipts of their own. A late receipt cancels unlaunched retries only, not packets already emitted. A lost receipt does not undo a delivery. Any failure notice sent back from a remote node must itself travel.

Connection setup and data

A session is set up along the pinned route and its reverse with three messages: SYN (request), SYN-ACK (reply), and ACK (acknowledgment).

- The initiator may send data after it receives SYN-ACK and queues the ACK.
- The receiver releases data to its application only after it receives the final ACK.
- An existing session may be reused. A new session needs a full handshake.
- Each peer connection has at most 64 unacknowledged data packets. Bytes are delivered to the application in order.
- Endpoints acknowledge received data. Duplicate data is never delivered twice within one session.

Endpoint timers and retries

For each endpoint attempt, T_0 is the sum of flight, serialization, and relay processing times along the pinned route, using the geometry at the moment the packet is enqueued, with moving receivers, empty queues, and no waiting for closures. If a link is blocked at that moment, use pure light time for it as if it were open. Recompute T_0 when the packet is enqueued and set $R_e = 2 \times T_0 + 24$ hours. This is a timeout, not a delivery guarantee.

Counting attempts. An endpoint attempt is used when the packet is enqueued, even if a closure makes it wait. The next attempt is due when R_e expires, whether or not the earlier copy has launched. When a retry is enqueued, any earlier copy of the same data packet that has not yet launched is discarded, so a long closure does not build a backlog of duplicates. A hop attempt is used only when the packet is actually launched, as described above.

Message	Attempts	Notes
SYN	At most 4	A duplicate SYN may trigger another SYN-ACK, within that message's cap
SYN-ACK	At most 4	Triggers an ACK
Final ACK	No timer of its own	A duplicate SYN-ACK permits another ACK
Data packet	At most 4	A retry gets a new network packet ID and keeps its session byte sequence, so it can cross the relays again

Endpoint retries overlap hop retries on purpose: the endpoint timer does not wait for hop retries to run out. Every copy uses link capacity. The quota exemption covers only this automatic procedure, so an application resubmission or a new session counts as new traffic.

When the attempts run out, the local attempt ends with delivery status unknown. Late packets for an abandoned session are ignored. Application recovery and how work carries across sessions are your decisions.

Expiry and duplicate records

- Every network packet expires 30 days after it was first created, including all its hop and endpoint retries.
- Expiry stops forwarding. It never cancels a financial obligation.
- Packet IDs are kept for the packet's lifetime plus 30 days after its last receipt. Session identities are never reused.
- A session expires after seven days without receiving a session packet, or on a declared reset.
- You define bounded long-term records, recovery, and safe application resubmission.

Scheduled and additional disruptions

- Maintenance windows happen once: B to Neptune during hours [2, 26) and B to Ceres during hours [240, 264), both directions.
- A launch fails if its flight interval $[t_e, t_a]$ overlaps a maintenance window on that link. Serialization time is not counted. A blocked photon path cannot launch.

- Maintenance and geometry are known in advance, so a sender waits whenever it knows that a flight would overlap maintenance or a blocked path. Incidents are not announced: a launch emitted inside an incident window simply fails.
- Queues wait until a valid launch is possible, within the packet lifetime.
- A pinned route can wait, or you can replace it with a newly handshaken session.
- In each tested window, one additional incident may occur: a 72 h gateway isolation, a 6 h forced-loss incident, or an endpoint reset. Natural geometric closures are not incidents.
- **Your S2 incident.** In the S2 stress run (Section 8) you choose the incident yourself: its kind, the node it hits, and its start time. It sits on top of the scheduled maintenance windows. Choose the one that hurts your design most and explain why. It does not count as the one optional incident for other tested windows.

What you must trace

Every trace shows each launch, receipt, and retry that affects a financial state. The queue cap, the order of simultaneous arrivals, and the 64-packet window are part of the model. You only need to discuss them where your design comes near them, for example a large backlog after a long closure.

6. Your balance sheet and scenarios

All numbers here are synthetic. You declare the money and shares that exist, who holds them, and where. Then you run your design through scenarios you choose. Nothing is assigned. The limits below keep the pool closed and the results comparable.

Your opening balance sheet

Before tracing anything, publish a table with one row per account: account name, settlement, NeoDollars, and shares. It must follow these rules:

- The total across all accounts is at most \$500,000 (NeoDollars) and at most 5,000 shares.
- There are 4 to 10 named accounts, located at 3 or more different settlements. The accounts are fixed for the whole run. You may not invent extra identities to gain communication capacity or any other advantage.
- Institutions you charter, including operators, guarantors, and agents, start with nothing.
- Every account starts with only the assets you list, at the settlement you list. The hour-0 balances and locations are authoritative.
- Before hour 0 you may only set up communication sessions, and no earlier than hour -168. No financial action is allowed before hour 0. No maintenance window or incident applies before hour 0. Every such packet counts in the rolling quota windows and in your communication totals. Sessions still expire after 7 days without a session packet.
- All your scenarios use this same balance sheet. Each run starts from the full opening balance sheet (a reset), and you must declare every reset.

Choose it to test your design, not to make life easy. Putting accounts far apart, or holdings where they are hardest to use, is a good way to find weaknesses.

Your scenarios

A scenario is one run that starts from the opening balance sheet at hour 0 and ends at a time you state. Scenarios start at hour 0 because the maintenance windows happen at fixed hours. Only the shifted-epoch runs (E5) start later. You may run several scenarios together on one balance sheet, or run them separately with a declared reset. The price-rising and price-falling runs in item 3 below must be separate runs.

Together, your scenarios must cover all four of these:

1. **Moving value between settlements.** A trade, a transfer, or both, where the person who receives the result can only use it after two settlements agree. Use accounts that are far apart.
2. **An obligation that stays open.** A contract that stays open for at least 240 hours and ends by payment, maturity, or a funded default.
3. **A payment that depends on later prices.** At least one product whose payments depend on prices observed after the contract opens. The contract must name which observations count: only the final price, an average, a barrier, or another rule. Futures, options, and short positions are examples. Run it twice: once with a price path that rises and once with a price path that falls. You choose the prices. Each path must move at least 20% away from the opening price at some point.
4. **Two different kinds of product.** Show at least two different kinds of product from the scope list in Section 1, for example a share trade and a futures contract.

Section 8 also asks for one stress run with an incident you choose (S2).

Every scenario must end with every obligation in a stated final state: paid, matured, in funded default, or still open with a named, funded owner (Section 9).

Making your scenarios meaningful

A scenario that never strains your design proves little. So:

- **Scale.** In at least one scenario, at least 25% of your opening NeoDollars must be encumbered by open positions at the same moment.
- **Exposure.** For each scenario, report each side's largest possible loss (its exposure) and how much of your capital is encumbered (capital utilization).
- **A real constraint.** Show one variation of size or price path that makes a real constraint bind: a margin call, a default, or a suspension. Bounded products, such as capped payoffs, are allowed. If nothing binds, show why not.

Prices from outside

If a product needs a price from outside the exchange, name the source, the one settlement where it releases each observation, and how often it releases them. The source releases each signed observation as a single packet, locally at that one settlement only. There is no default remote broadcast. You decide whether, how, when, and by what route the observation reaches anyone else, including who sends it and which quota it uses. Section 4 (Classifying traffic) says which messages may use the direct service and which must use the backbone.

How payments are counted

For any contract that pays on prices, the total paid over the contract's life is fixed by its terms and the observations it names. You design interim marking, but all interim and final payments together must add up to that total. Do not count a variation payment and the final payout as two separate obligations.

Example: a cash-settled future on 10 contracts with multiplier 100 and entry price 100, whose payment depends only on the final price P , pays the long side $10 \times 100 \times (P - 100)$ NeoDollars in total, and the short side the opposite.

When is something complete?

These definitions apply to every scenario and to every completion time you report. Times run from the start of the scenario.

- **Trade or transfer complete.** Your authoritative record shows it final, each party can use the result as a spendable balance at their own settlement, and the matching debits are recorded. The completion time is the later of the moments when each party can use the result.
- **Position open.** Both sides' margin is encumbered and the position is recorded at the clearing location's authoritative ledger.
- **Position settled.** Report three moments separately. (1) Discharge: the total amount owed, with interim and final payments counted once, is extinguished on your authoritative record. (2) Backed claim: the winner holds a recorded claim backed by assets already encumbered. (3) Spendable: the winner can use the amount as a balance at the winner's own settlement, and the loser's encumbrance is debited or released to match. Latency credit uses the spendable moment. If your design only ever gives a backed claim, say so. You may report its time, but it is scored as not yet spendable.
- **Funded default.** The loser's obligation is declared unpaid under your margin rule. The winner is paid, or holds a recorded claim, entirely from assets already posted and encumbered. No new assets are created and nothing is left unbacked. You define where the authoritative record sits. Report the backed-claim moment and the spendable moment, as for any settlement.

Products

Specify each product's quantity, unit, location, maturity, pricing inputs and valuation source, who bears losses, and what happens at end of life. Take all funding from your opening balance sheet. If you support a currency, it is a unit issued by one of your institutions and fully backed by assets that are encumbered for that purpose. It never creates free backing.

Access at every settlement

Every listed settlement must support local participation and access to your services, including at least one product with a payment that depends on later prices. You may impose justified product or risk restrictions. You may not solve this by permanently excluding a settlement, removing price-dependent products, or halting all services indefinitely.

You show this with comparable funded clients at other settlements (Section 8). Each demonstration is a re-run from reset in which one of your accounts, with its holdings, sits at another settlement. It never adds capital.

7. Funding, margin, and probability rules

The funding pool is closed, one margin rule must work for both price directions, and every reported time carries its probability. Your design must satisfy these three rules first.

Conserve all funding

- An asset supports only one use at a time.
- Every guarantee, credit advance, insurance payout, liquidity commitment, and unit of collateral must come from your opening balance sheet through an explicit, valid transaction. No institution can declare new wealth, outside backing, or hidden reserves.
- A loan or a new currency creates matching claims and liabilities. It does not create free backing.
- Record every funding source with its owner, location, liability, and encumbrance.
- Count assets committed by clients, remote agents, guarantors, and institutions in the same capital measures, even if they are held outside an exchange.
- An account at an agent does not put any cash or shares at that agent. Assets move only under your submitted procedures and the communication rules. A publication time passing does not by itself make any amount payable somewhere else.
- After hour 0, fees and approved capital contributions may fund institutions. They debit the contributing accounts and take time wherever communication is needed.
- Do not double-count collateral, sale proceeds, loan claims, or guarantees as separate available resources.

One margin rule for both price directions

State one margin, valuation, and default procedure before tracing either price path. It may use:

- contract terms;
- current and earlier received observations;
- positions;
- risk assumptions you declare.

It may not refer to future prices, to which run it is, to who will end up paying, or to a report that has not been published yet. Your price paths are test scripts, not advance information. Everything you do before the first price observation after the contract opens must be the same in both runs whenever the local information is the same.

After an observation is received, your rules may respond to it. Margins may differ by direction if you justify the difference by risk. Report the two runs separately. Scoring weights them equally.

Probability and timing

The no-loss trace shows the timeline when every packet arrives. It is conditional evidence, not a promise.

Every reported time that depends on the direct service must include:

- the probability that all necessary direct packets and copies arrive by that time under the direct loss formula;

- what happens if they do not arrive: either a fallback with its timing, or a clear statement that the operation stays incomplete.

Times that depend on the backbone need a lighter standard. Give the conditional trace, the loss probability of each launch on the route, and the probability that a hop uses all four of its launches and is abandoned. Exact values or bounds are fine, and you must say which you give. Also say what happens to the financial state if a hop is abandoned.

Count dependencies correctly. For n independent copies that each fail with probability p , all fail with probability p^n , so success is $1 - p^n$. Copies are not independent if they can share a closure, a forced-loss window, or a queue. Do not multiply probabilities of events that are not independent. Check departure times, solar visibility, quotas, and serialization before you accept a claim that something succeeds by a deadline.

Latency credit considers the probability and the consequences of completion, not just the fastest conditional arrival. No conditional time counts as certain. Because of random loss, no unconditional delivery deadline exists.

Complete traffic accounting

Communication efficiency counts every packet:

- client, agent, and price-source direct transmissions, with their acknowledgments and copies;
- every backbone hop, control message, retry, and receipt.

Report direct and backbone totals separately and together. Direct traffic uses each principal's own quota, not the shared 600-packet quota. Moving traffic to a different service never removes it from the efficiency score.

Compare alternatives under the same funding, price paths, communication assumptions, and guarantees. A faster or cheaper result is not better if it weakens a guarantee or excludes users.

How the measures are defined

- **Peak encumbered assets.** The largest amount encumbered at one moment, reported separately for NeoDollars and for each kind of share.
- **Asset-hours.** For each asset type separately, the amount encumbered added up over time, in units $\times$ hours. NeoDollars and shares are never added together.
- **Value settled.** The NeoDollar value of the trades, transfers, and payments completed in a scenario. A trade counts price $\times$ quantity. A transfer counts the value moved at your stated prices. A price-dependent contract counts its notional (quantity $\times$ multiplier $\times$ opening price).
- **Capital efficiency.** Peak encumbered NeoDollars divided by NeoDollar value settled, alongside the asset-hours. Lower is better only if every guarantee is fully funded.
- **Communication efficiency.** Total packets (direct plus backbone) divided by the number of completed transactions in the scenario. Lower is better.

8. Evidence you must submit

Submit eight items: three about service (S1 to S3) and five about calculations (E1 to E5). Together they fit in a 12-page appendix. Code is optional. Calculations, spreadsheets, plots, pseudocode, and written

reasoning are all acceptable. Every important claim must be understandable from the submitted documents alone.

Service evidence

Item	What to show
S1. Scenario traces	A conditional (no-loss) trace at the epoch for each of your scenarios, with maintenance and geometry applied. Together they must include the value move, the long-lived obligation, the price-rising run, and the price-falling run (Section 6). For each trace report the completion time, all communications, peak encumbered assets, and asset-hours by asset. For each scenario also report exposure and capital utilization, and show that your scale requirement is met in at least one scenario. Add the one variation that makes a real constraint bind (Section 6). There are no fixed deadlines: report your own times, with their probabilities (Section 7). If several scenarios share one run, the capital measures include the encumbrances from all of them.
S2. Stress and recovery	Take one scenario that includes a price-dependent product and add one incident (Section 5). Choose the kind, the node, and the start time that hurt your design most, and explain why. Report the duration of lost service, the state of every obligation during the incident, and the recovery after connectivity returns. Recovery means financial service has resumed, not that a packet was received. Maintenance comparison: repeat the run with scheduled maintenance removed (natural geometry stays) and compare. Alternative: make one change to your design, or run one sensitivity check, under identical funding and guarantees. A stressed run may pause or reject activity if you explain the obligations and the recovery.
S3. Access at every settlement	A table with one row for each of the nine settlements. For each transaction type you support, give the best route from that settlement to the authorities the transaction needs (for example the clearing location, a ledger, or a price source), the one-way delay, and the availability. Give the table at hour 0 and again at hour 300, so the effect of motion shows. Availability at hour h is the fraction of the 24 hours starting at h in which every link on the route can launch. Then give a full no-loss timeline of one of your value-move scenarios for a comparable funded client at three settlements: the best, the median, and the worst by your own delay ranking. Each is a re-run from reset (Section 6).

Every trace uses one row per step. The example row below is illustrative only. Its numbers are not real results.

Time	Actor	Local knowledge	Action	Packet or transmission	Arrival	Financial state after
Hour or timestamp	Who acts	What that actor knows at this moment	What it does	Direct or backbone, link, and launch time	Actual arrival time, or loss	Balances, encumbrances, and obligations
Example: h 0.0003	Earth clearing service	Has the buyer's order and margin. Knows nothing about Mars.	Sends SYN toward Mars	Backbone, Earth to Relay A, launched at h 0.0003	Example: h 0.31	Buyer's \$10,000 encumbered at Earth. No obligation yet.

Label every conditional example and every probability calculation clearly. State which claims depend on assumptions.

Calculation evidence

Item	What to show
E1. Orbital calculations	Your propagation method, frame, time units, numerical tolerances, and light-time solution. Your epoch positions checked against the table in Section 3. Worked positions at a later time. One worked example where receiver motion changes the arrival time. Forward and return messages analyzed separately.
E2. Probability and traffic	Every success-by-deadline probability for the direct service, with dependencies handled correctly and the outcome when packets do not arrive. Backbone launch and hop-abandonment probabilities for the routes you use (Section 7). Packet counts for direct, backbone, and total. Quota use against both quotas. Link-capacity checks.
E3. Balances and capital	Balances and obligations over time for every scenario, with the source, owner, location, liability, and encumbrance for each asset. A check that funding is conserved and that no asset serves two uses at once.
E4. Long-horizon scan	Tier 1 (1 of 3 points): a scan of at least 200 Julian years of model time, with the step size stated and an argument for the shortest closure it could miss. Tier 2 (2 of 3): Tier 1 plus one refined event boundary, with its bracket, tolerance, and the effect on packets in flight. Tier 3 (3 of 3): Tier 2 plus link availability and route-delay ranges for every settlement, including the poorest service you found. The poorest service is the settlement and window with the lowest availability or the longest delay. An analytical bound may replace part of the scan if it is stated clearly.
E5. Shifted epochs	See the next subsection. Scored in tiers.

Tiers for E4 and E5. Tiers are cumulative: each tier includes the one before it, and each earns the points shown. Extra runs beyond the tier you claim earn nothing more, so state which tier you are claiming. A monthly sample alone cannot show that no shorter outage exists.

Shifted epochs (E5)

An offset of k years starts your scenarios at $t = k \times 365.25$ days. Advance all planets and both relays from the original epoch. Reset only the financial balances, back to your opening balance sheet. Do not reset orbital phase, and do not replay the original maintenance windows or your S2 incident. Observation and maturity times shift with the scenario.

- **Tier 1 (1 of 3 points): offsets of 1, 10, and 100 Julian years.** One summary table. For each offset, list the delays and availability on the routes your scenarios use, then give a conditional completion time or a justified suspension.
- **Tier 2 (2 of 3): one difficult epoch you choose.** Add a value-move scenario and the price run that is harder for your design at that epoch, with the reason you picked it.
- **Tier 3 (3 of 3): both price directions.** Add the other price run at that epoch, or explain why your design is symmetric enough that it needs no separate run, and give the funded consequences of any suspension.

Do not treat any early completion time as a promise of connectivity at later epochs.

Suggested page budget

Document	Contents	Pages
Evidence appendix	E1 orbital calculations	1.5
	S1 and E2 to E3: traces, probabilities, balances	5
	S2 stress and recovery	2
	S3 access table and timelines	1
	E4 long-horizon scan	1.5
	E5 shifted epochs	1
Design paper	Architecture and operating rulebook	5
	Guarantees, conditions, and costs (including the service table)	2
	Risks you discovered and how you treat them	2
	Long-term operation (Section 9)	2
	Deployment assessment (Section 9)	1

Use type of at least 10 pt and margins of at least 1 inch. You may attach machine-readable traces (for example a spreadsheet or a log file) as supporting material only. The appendix must stand on its own without them.

9. Continuing operation and deployment

Your market has no planned end. Your scenarios, the 30-day packet lifetime, and the 200-year orbital span are test windows, and none of them ends the market. No asset, queue, debt, quota, or identifier resets automatically when a window closes.

Rules that keep working

Explain why your procedures stay valid at any model epoch, including:

- a long period with no usable route, and the restoration afterward;
- a backlog of work after a route reopens;
- an obligation that matures after your longest worked example;
- an old transport packet that expires while a financial commitment is still unresolved.

State a conditional service promise wherever geometry cannot support an unconditional one. Numerical sampling is evidence, not proof of an all-time guarantee, so say which claims rest on the model and which rest on your rules.

Show how each of these continues without a reset:

- order and contract lifecycles, including maturity processing;
- capital replenishment through earned or contributed resources, with no free future endowment;
- bounded storage and identifier use;
- quota enforcement on both services;
- the infrastructure you assume.

For storage and identifiers, you may state an engineering lifetime (for example, identifiers sized to last a stated number of years) or an explicit identifier-growth assumption. Say what happens when the limit is reached. Session identities are still never reused within that lifetime.

No obligation is cleared automatically. Every unresolved obligation keeps a funded owner until your rules resolve it.

Support recurring work by declaring a demand rate you accept, and say what happens to load above it. A system with finite capacity cannot serve unlimited demand, and you should not promise that it can.

Deployment assessment

The baseline leaves out real effects on purpose. Identify the extra effects you think matter and say whether each would change your financial guarantees. Give a numeric estimate, with sources and assumptions, for at least two. Candidates include local rotation and ground visibility, Doppler shift, local satellite handoff, planetary shadow, and timekeeping.

Effect	Size estimate	Source and assumptions	Effect on your guarantees
Your first effect			
Your second effect			

Higher fidelity is welcome, but baseline comparisons must stay reproducible from the common data. You do not need to run an exact relativistic solar-system simulation, and you should not invent orbital parameters for local satellites that the brief does not give. Say plainly where your assumptions stop. Honest model limits are good engineering judgment and are not penalized.

10. Submission and scoring

You submit a design paper and an evidence appendix by 10:00 AM Eastern on Sunday, October 4, 2026, which is 40 hours after the 6:00 PM Eastern start on Friday, October 2. The deadline is real clock time and has nothing to do with simulation hours. Your submission is scored out of 100 points. Correctness and asset protection carry the most weight.

What to submit by the deadline

Deliverable	Limit	Contents
Design paper	12 pages including diagrams and references	Architecture, operating rulebook, supported products, responsibilities, your guarantees and their conditions, the risks you found, and your limits. Include a service table: for each product and settlement, what is offered, under what conditions, and with what restrictions. Keep exact procedures separate from future work. There is no live questioning, so state what each actor knows, does, and owes when a message is lost, delayed, duplicated, or contradicted.
Evidence appendix	12 pages	Items S1 to S3 and E1 to E5 from Section 8

The paper and the appendix are what get scored. Optional code or spreadsheets may come with them, but every important claim must be understandable from the documents alone. Follow the hackathon-wide rules on sources and assistance.

Scoring

Criterion	Points	What earns credit
Correctness and asset protection	30	Your financial state is exact and consistent, and your guarantees can be enforced. At the top of the range, authority, states, obligations, and recovery are all stated exactly.
Economic usefulness	25	Access and completion time 10, communication efficiency 5, capital efficiency 5, recovery 5. Timing is judged against the guarantees you give and the probability of completion, not raw speed. Capital and communication efficiency use the definitions in Section 7. Lower capital earns nothing if it comes from unfunded exposure. Recovery means financial service has resumed, not that a packet was received. The price-rising and price-falling runs carry equal weight. There are no fixed deadlines to hit.

Criterion	Points	What earns credit
Risk discovery and response	20	Original identification, ranking, and treatment of weaknesses. Interactions between your own rules and honest limits score highest. Your own scenarios count here: judges look at how hard they push on your design, whether a real constraint binds, and whether you chose the incident and its timing to hurt it.
Quantitative evidence	15	Three points for each of E1 to E5, for correct, reproducible calculations and sensitivity analysis. A missing item scores zero for its three points.
Clarity and reproducibility	10	A reader can trace every important claim from the documents alone, and proofs, assumptions, and limitations are kept separate.

Scoring anchors

Judges use these anchors to place a submission within each range.

Criterion	Top of the range	Middle	Bottom
Correctness and asset protection	Every balance, encumbrance, and authority statement agrees across the paper and the traces. Each guarantee names its conditions. Recovery is exact.	Mostly consistent, with a few gaps or unstated conditions.	Balances do not reconcile, an asset is used twice, or a guarantee has nothing enforcing it.
Economic usefulness	All nine settlements are served, with restrictions stated. Times come with probabilities and fallbacks. Packet counts and capital use are low for the guarantees given. Recovery restores financial service.	Access is stated but thin, or times are conditional only.	Settlements are excluded, or speed comes from weakening a guarantee.
Risk discovery and response	Original risks, ranked, with tests that really bind and a treatment for each. Interactions between the team's own rules are found.	Standard risks listed, with treatments.	Risks are generic or untested.
Quantitative evidence	Correct and reproducible work at the tier claimed, with sensitivity analysis.	Correct at a lower tier, or errors that do not change conclusions.	Missing, or errors that change conclusions.
Clarity and reproducibility	A reader can trace every claim. Proof, assumption, and limitation are labeled.	Traceable with effort.	Claims depend on missing detail.

Technology names cannot replace procedures. Words such as blockchain, consensus, auction, or clearinghouse earn nothing on their own.

Submission checklist

- Design paper (12 pages), including the service table.

- Evidence appendix (12 pages): S1, S2, S3, E1, E2, E3, E4, E5, with the tiers you claim for E4 and E5.
- Your opening balance sheet, with 4 to 10 accounts at 3 or more settlements and totals within the limits.
- Scenarios that cover the value move, the obligation open for 240 hours or more, the price-rising and price-falling runs, and two kinds of product.
- One margin rule stated before either price path is traced.
- The scale requirement met, exposure and capital utilization reported, and one variation that makes a real constraint bind.
- An S2 incident you chose, with your reason.
- Direct-service probabilities and backbone hop probabilities, with fallbacks.
- Spendable, backed-claim, and discharge moments reported separately for every settled position.
- Type at least 10 pt, margins at least 1 inch.

11. Clarifications during the event

If something in this brief is unclear, or two parts of it disagree, ask the organizers. Answers are posted as numbered, dated clarifications in the same place where this brief was released. Where a clarification and this text disagree, the clarification governs. Teams are responsible for reading every clarification that is posted.